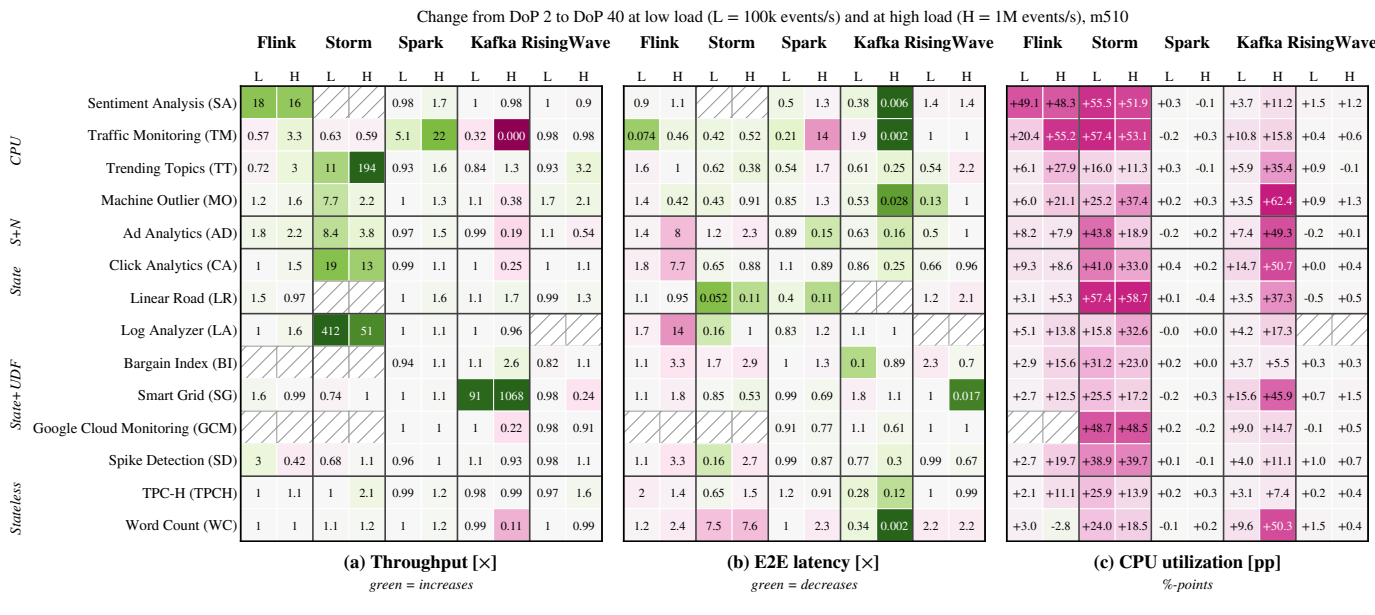
Change from DoP 2 to DoP 40 at low load (L = 100k events/s) and at high load (H = 1M events/s), m510
Flink
Storm
Spark
Kafka
RisingWave
L
H
CPU
State
S+N
State+UDF
Stateless
Sentiment Analysis (SA)
Traffic Monitoring (TM)
Trending Topics (TT)
Machine Outlier (MO)
Ad Analytics (AD)
Click Analytics (CA)
Linear Road (LR)
Log Analyzer (LA)
Bargain Index (BI)
Smart Grid (SG)
Google Cloud Monitoring (GCM)
Spike Detection (SD)
TPC-H (TPCH)
Word Count (WC)
18 16 0.98 1.7 1 0.98 1 0.9
0.57 3.3 0.63 0.59 5.1 22 0.32 0.000 0.98 0.98
0.72 3 11 194 0.93 1.6 0.84 1.3 0.93 3.2
1.2 1.6 7.7 2.2 1 1.3 1.1 0.38 1.7 2.1
1.8 2.2 8.4 3.8 0.97 1.5 0.99 0.19 1.1 0.54
1 1.5 19 13 0.99 1.1 1 0.25 1 1.1
1.5 0.97 1 1.6 1.1 1.7 0.99 1.3
1 1.6 412 51 1 1.1 1 0.96
0.94 1.1 1.1 2.6 0.82 1.1
1.6 0.99 0.74 1 1 1.1 91 1068 0.98 0.24
1 1 1 0.22 0.98 0.91
3 0.42 0.68 1.1 0.96 1 1.1 0.93 0.98 1.1
1 1.1 1 2.1 0.99 1.2 0.98 0.99 0.97 1.6
1 1 1.1 1.2 1 1.2 0.99 0.11 1 0.99
0.9 1.1 0.5 1.3 0.38 0.006 1.4 1.4
0.074 0.46 0.42 0.52 0.21 14 1.9 0.002 1 1
1.6 1 0.62 0.38 0.54 1.7 0.61 0.25 0.54 2.2
1.4 0.42 0.43 0.91 0.85 1.3 0.53 0.028 0.13 1
1.4 8 1.2 2.3 0.89 0.15 0.63 0.16 0.5 1
1.8 7.7 0.65 0.88 1.1 0.89 0.86 0.25 0.66 0.96
1.1 0.95 0.052 0.11 0.4 0.11 1.2 2.1
1.7 14 0.16 1 0.83 1.2 1.1 1
1.1 3.3 1.7 2.9 1 1.3 0.1 0.89 2.3 0.7
1.1 1.8 0.85 0.53 0.99 0.69 1.8 1.1 1 0.017
0.91 0.77 1.1 0.61 1 1
1.1 3.3 0.16 2.7 0.99 0.87 0.77 0.3 0.99 0.67
2 1.4 0.65 1.5 1.2 0.91 0.28 0.12 1 0.99
1.2 2.4 7.5 7.6 1 2.3 0.34 0.002 2.2 2.2
+49.1 +48.3 +55.5 +51.9 +0.3 -0.1 +3.7 +11.2 +1.5 +1.2
+20.4 +55.2 +57.4 +53.1 -0.2 +0.3 +10.8 +15.8 +0.4 +0.6
+6.1 +27.9 +16.0 +11.3 +0.3 -0.1 +5.9 +35.4 +0.9 -0.1
+6.0 +21.1 +25.2 +37.4 +0.2 +0.3 +3.5 +62.4 +0.9 +1.3
+8.2 +7.9 +43.8 +18.9 -0.2 +0.2 +7.4 +49.3 -0.2 +0.1
+9.3 +8.6 +41.0 +33.0 +0.4 +0.2 +14.7 +50.7 +0.0 +0.4
+3.1 +5.3 +57.4 +58.7 +0.1 -0.4 +3.5 +37.3 -0.5 +0.5
+5.1 +13.8 +15.8 +32.6 -0.0 +0.0 +4.2 +17.3
+2.9 +15.6 +31.2 +23.0 +0.2 +0.0 +3.7 +5.5 +0.3 +0.3
+2.7 +12.5 +25.5 +17.2 -0.2 +0.3 +15.6 +45.9 +0.7 +1.5
+48.7 +48.5 +0.2 -0.2 +9.0 +14.7 -0.1 +0.5
+2.7 +19.7 +38.9 +39.7 +0.1 -0.1 +4.0 +11.1 +1.0 +0.7
+2.1 +11.1 +25.9 +13.9 +0.2 +0.3 +3.1 +7.4 +0.2 +0.4
+3.0 -2.8 +24.0 +18.5 -0.1 +0.2 +9.6 +50.3 +1.5 +0.4
(a) Throughput [×]
green = increases
(b) E2E latency [×]
green = decreases
(c) CPU utilization [pp]
%-points